Heart Disease Prediction using Medical Data

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Initial Contextual Report

COMP1682 Final Year Project Proposal

BSc (Hons) COMP-1682-Final Year Project

Word Count: 4462

Abstract

The study examines predictive modelling using machine learning (ML) and deep learning (DL), particularly focusing on the opacity in predictive modelling within the healthcare domain. The study attempts to analyse the prediction of heart disease using the various approaches of predictive modelling. The study is comparative in the sense that it attempts to evaluate the models of Logistic Regression, Random Forest, XGBoost, and an Artificial Neural Network (ANN), and combine them with the explainable artificial intelligence (XAI) models of SHAP and LIME. The dataset is prepared, predictive models created, and the literature for the study reviewed. The most explainable model to integrate is presumed to be the one for the web-based decision support system for interaction with the user in real-time.

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my project supervisors for their guidance, feedback, and academic support throughout the development of this report. I also acknowledge the resources and learning support provided by the University, which contributed to the successful completion of this work.

Table of Contents

Acknowledgements 2
Table of Contents 3
1. 1 Introduction 4
   1.1. Background 4
   1.2. Aim and Objectives 4
      1.2.1. Aim 4
      1.2.2. Objectives 5
   1.3. Approach 6
2. Literature Review 6
   2.1. Approach to Literature Searching 6
   2.2. Identifying the Problem 7
   2.3. ML Approaches 7
   2.4. XAI in Healthcare 7
   2.5. Conclusions 8
3. Project Research 8
4. Legal, Social, Ethical and Professional Issues and Considerations 9
5. Requirements 10
   5.1. Analysis of Requirements 10
   5.2. Comparisons of Systems 11
   5.3. Requirement Elicitation 11
   5.4. Functional Requirements 12
   5.5. Non-functional Requirements 12
6. Design 13
7. Prototype 14
   7.1. Prototype Approach 14
   7.2. Review of Technology 14
   7.3. Development Progress 15
   7.4. Reflection 15
8. Conclusion 16
9. References 17

1. Introduction

1.1. Background

The leading cause of global mortality is Cardiovascular diseases (CVD) which dramatically impacts international healthcare systems. Most heart-related illnesses develop over time, which means identifying risk factors before serious damage occurs is critical. Preventative measures often rely on clinical judgement, imaging, and labs which can be efficient, but are often time costly, and even more difficult to scale as complex patient data is involved.

As more data becomes available, phenomena such as Machine Learning (ML) and Deep Learning (DL) have been applied in the prediction of CVD. When predicting risk of CVD, patient information such as age, cholesterol, blood pressure and lifestyle data the fore-mentioned techniques outperformed conventional statistical algorithms, especially in terms of complex and non-linear data.

Regardless of the mentioned benefits, implementing machine learning models into practice is still not commonplace. The main reason is the absence of explainability when it comes to more developed models, particularly the ensemble and neural network approaches. Delivering healthcare predictions that cannot be explained is inherently untrustable, and performance does not matter. Aside from this, when the majority of studies focusing on prediction models, the only thing they tend to be concerned with is how accurately the prediction can be made. This is where explainable prediction models that can be practically deployed come into play and is the motivation for the development of a web-based decision-support prototype with a focus on transparency, usability, and ethics.

1.2. Aim and Objectives

1.2.1. Aim

The aim of this project is to design and evaluate an explainable machine learning based system for the early prediction of cardiovascular disease and to integrate the most suitable predictive model into a web-based decision-support application that enables real-time user interaction.

1.2.2. Objectives

In the interest of clarity, feasibility, and alignment with ethical and practical concerns, the objectives of this project have been revised from the initial proposal. Revised objectives are stated as follows.

i. To conduct a critical review and analysis of the literature on the prediction of cardiovascular diseases using the techniques of machine learning, deep learning, and explainable artificial intelligence, with a view to determining the limitations and gaps in the existing body of research.

ii. To support reliable and reproducible model training by preprocessing the publicly available heart disease dataset, by means of class balancing, data cleaning, and feature scaling.

iii. To create and assess a number of predictive models, namely Logistic Regression, Random Forest, XGBoost, and a feed-forward Artificial Neural Network, using the same assessment criteria so as to facilitate an equitable assessment.

iv. To implement explainable artificial intelligence methods, in particular, SHAP (Lundberg and Lee, 2017) and LIME (Ribeiro et al., 2016), in order to provide transparency and trust in healthcare decision support systems, by providing global and local explanations of model predictions.

v. To embed the best predictive model, selected on the basis of performance and explainability, into a comprehensive, full-stack web-based application, which will enable users to enter data on a patient and receive predictive results in real time.

vi. Focus on maintaining ethical professional practice at all stages, including adherence to the GDPR (European Union, 2018) and the institution's

research ethics guidelines, particularly concerning responsible and reproducible AI.

1.3. Approach

This research formulates a systematic explainable framework for predicting heart disease. The research uses a heart disease dataset accessible from Kaggle (Damarla, 2024). The dataset includes features of patients and is raised to assess the patients' cardiovascular risks. The dataset also allows for reproducible research and comparative analysis.

The research assesses the framework, which includes the stages of data preprocessing, cleaning, feature scaling, and class balancing, to assist in model training. The framework includes multiple predictive models, including the baseline Logistic Regression and advanced machine learning and deep learning models such as Random Forest, XGBoost, and Artificial Neural Network. All models will be evaluated using the same performance metrics.

Explainable artificial intelligence (AI) is used to solve the problem of the predictive models in the healthcare domain lacking transparency. The global feature explanation for AI is used with SHAP (Lundberg and Lee, 2017), and LIME (Ribeiro et al., 2016) is used for local model explanation. The most optimal model is based on predictive performance and explainability.

The last model is embedded into a full stack web based application that allows users to input data on the patient and receive predictions in an explainable machine learning, and becomes a proof of concept for a real world, user-friendly decision support system to go beyond academic evaluation.

2. Literature Review

2.1. Approach to Literature Searching

I utilized institutional access to major academic databases for the literature review including IEEE Xplore, SpringerLink, ScienceDirect, PubMed, and Google Scholar. Attention was given to peer-reviewed journals and conference publications from the last decade. Among the studies identified using the search terms “heart disease

prediction", "machine learning and cardiovascular disease", "explainable artificial intelligence", "SHAP", and "LIME"", only those relevant to the prediction of healthcare outcomes, clear methodologies, and insight to either predictive performance or interpretability of the models were selected.

2.2. Identifying the Problem

Most studies in reporting the prediction of cardiovascular disease using machine learning focus on predictive performance, as many studies providing improved accuracy through ensemble and deep learning models. However, these methods tend to lack predictive power, leading to opacity and a lack of trust in prediction in a clinical setting.

Moreover, researchers often work with very small, benchmark data sets, and pay little attention to the reproducibility, data set bias, and generalisability of their work. Most studies are done in very controlled experimental settings with little regard for actual performance, use, and human interaction. These issues reinforce the need for a predictive system that provides a practical trade-off between performance and explainability.

2.3. ML Approaches

Techniques like Logistic Regression, Decision Trees, Random Forest, and gradient-boosted models are popular for predicting heart disease. Because of their simplicity, linear models are usually set as baseline models. However, due to their ability to model complex relationships between clinical features, tree-based and boosting methods are usually preferred and are believed to outperform standard statistical methods.

Despite these purported gains, the majority of the studies are still focused on performance. There is little to no attention paid to the models' robustness, or the explanations on the individual patient level, which limits the appropriateness of these methods for accurate and dependable decision support systems for healthcare.

2.4. XAI in Healthcare

The need for transparency and accountability in predictive systems has enabled the field of healthcare machine learning to evolve with Explainable Artificial Intelligence (XAI). In clinical decision-support frameworks, predictive models are

only actionable to the extent that they are supplemented with explanatory features that facilitate the understanding and evaluation of model output by the target end-user clinicians.

Literature discusses model-agnostic techniques such as SHAP (Lundberg and Lee, 2017) and LIME (Ribeiro et al., 2016) for global and local model explanations; however, as in other domains, the necessary explanations for health care systems are treated as secondary, and the usefulness of explanations remains unjustifiably low. This highlights the necessity of inter-system communication for explainability in healthcare predictive systems.

2.5. Conclusions

Research shows that machine learning methods can predict cardiovascular diseases more accurately than older statistical methods. Still, most research is focused on performance, and explainability, reproducibility, and the practicality of the research is ignored. Such advanced techniques as deep learning and ensemble methods are often considered black-box models. This is problematic in clinical decision support systems where transparency is needed.

Additionally, most research relies on limited experimental designs with little explanation on deployment and end-user. Such imbalances warrant the need for predictive systems that are accurate and interpretable, as well as implementable. This review defines the focus on user-centered design and explainable artificial intelligence as a starting point for assessing the current systems and designing a predict heart disease system that is explainable and deployable.

3. Product Research

Product research studies available heart disease prediction and healthcare decision-support systems and predictive technologies. In contrast with the literature review, which is concerned with the theory, this section analyses real-world systems with regard to practical limitations on usability, transparency, and deployment.

By way of analogy, existing platforms for cardiovascular risk assessment have been analyzed based on some of the same criteria. These platforms have been analysed in regard to predictive capability, amount and type of explanation, and real-world usability,

which is non-technical. The selection of these criteria speaks to the technical and practical goals of the systems in the context of healthcare decision support.

A significant number of the existing systems use statistical scoring or basic prediction models. While the systems may be user-friendly and easily implemented, the systems frequently yield static risk assessments without elaboration, making the assessments' transparency and trustworthiness to be counter-explainable.

Some of the more sophisticated systems use machine learning. These systems do, however, tend to be self-contained and do not allow insight into the models' mechanics. Explanations for the decisions made are often merely generic and fail to be sufficient for any one specific case. The reasoning behind the assessments remains inadequate, and therefore, decision support is rendered ineffective.

Furthermore, numerous already available solutions are built for different types of health-care providers and require specialized knowledge and skills for output interpretation. Also, due to complex interfaces, and their limited support to non-technical users, systems are less likely to be adopted. The current systems, when optimized for usability and deployment, tend to predict either performance or accessibility not both.

These findings highlight the most significant gap in current systems: the lack of predictive systems with sufficient explainability and usability. This explains the system's requirements and design and anticipates combining explainable machine learning with a web interface to provide users with a system to calculate and explain cardiovascular risk in an accessible and transparent manner.

4. Legal, Social, Ethical and Professional Issues and Considerations

Predictive systems applied to medical decision support raise numerous concerns, in regard to the application of machine learning in the field of health care. since the project deals with health related data, the project must comply with the data protection laws. The dataset used is publicly available, doesn't contain personal information, and is fully compliant with the GDPR (European Union, 2018) law in means of data processing, transparency, and data protection.

Bias and the possible misinterpretation of the predictive models has been the subject of multiple ethical concerns. To counter, the system is constructed to be a decision support system to remove use as an automated diagnostic tool. This is designed to allow the prediction to support the clinician's judgement, reinforce the prediction with the model, and remove the possibility of automated decision making. The use of explainable artificial intelligence promotes transparency and accountability to model predictions and lowers the concerns of the users in regard to black box predictions.

The project is guided from a professional standpoint by the principles of research ethics by institutions, as well as the principles of ethics in the field of health care, and the guidelines of the field of predictive model development. The development of the project demonstrates consistency with standards related to reproducible and responsible Artificial Intelligence, as well as proper attribution.

The UK Data Protection Act 2018 was evaluated for legal applicability and found to be reinforcing requirements already addressed by the GDPR. Regulations regarding medical devices were also considered, but the system design was excluded as it does not provide any diagnosis or treatment recommendations, and therefore does not constitute a regulated medical device.

5. Requirements

This part describes the heart disease prediction system's requirement. The system's requirements are based on existing products, ethical and legal requirements, and the defined project scope. The requirements give the design and prototype development a framework.

5.1. Analysis of Requirements

This project is guided by requirements based on previous work regarding heart disease prediction systems and previous work's ethical and professional limitations. Current solutions focus on predictive power while ignoring transparency, responsible use, and systems that are easy to use.

Therefore, the system being designed needs to have accurate predictions and provide output that is explainable and ethically defensible. Explainable models are a requirement and not a nice to have. Furthermore, the system is not allowed to make

any clinical diagnosis and must function exclusively as a decision support tool with the use of de-identified data.

The system is also required to have a simple web-based architecture that allows the user to enter the patient's data and receive a prediction, as well as, a prediction guide. These restrictions limit the functions and features that are described in the requirements below.

5.2. Comparisons of Systems

System requirements were derived by comparing existing heart disease prediction systems and support systems for healthcare decision-making. The comparison criteria was based on product research and included prediction capability, explainability, usability and deployment suitability. Tools that are accessible, are often simplistic, while more advanced systems based on machine learning, improve prediction, but compromise interpretability and user friendliness.

This comparison serves to outline requirements for the system. In this case, the system must provide predictive answers while being able to explain the answers, and must also provide a user interface that is accessible to those that are not technically inclined. These observations directly inform the functional and non-functional requirements defined in the following subsections.

5.3. Requirement Elicitation

The project scope did not contain stakeholder analysis, and the requirements could be obtained via secondary sources, including literature, since there was no access to clinical stakeholders. Given the project`s academic nature, stakeholder engagement was eluded, and thus, the requirements were obtained through a literature review, analysis of other similar systems, and consideration of related legal, ethical, and other professional issues.

The analysis of the existing systems for the prediction of heart disease and the findings from the product research shed light on the issues of non-explainability, poor usability, and lack of deployment. These system limitations, combined with legal and ethical constraints, including GDPR and the ethics of using a prediction

system model, define the system requirements. Furthermore, the systems functionality order was also defined by the proposal project objectives.

This enables the requirements to be justified, realistic, and aligned with the purpose of the system which is to be a decision-support system and not a diagnostic clinical system.

5.4. Functional Requirements

The functional requirements define what the proposed system must do in order to meet the project objectives and address the limitations identified in existing systems.

1. The system shall allow users to input patient-related attributes relevant to cardiovascular risk assessment through a web-based interface.
2. The system shall process user-provided input data and generate a heart disease risk prediction using a trained machine learning model.
3. The system shall provide a clear prediction output indicating the likelihood of cardiovascular disease.
4. The system shall generate explainable outputs alongside predictions, providing insight into the factors that influence model decisions.
5. The system shall support both global and local explanations of predictions using explainable artificial intelligence techniques.
6. The system shall allow predictions to be generated in real time without requiring technical knowledge from the user.
7. The system shall operate using anonymised data and shall not store or process personally identifiable information.

5.5. Non-functional Requirements

Non-functional requirements describe the system's operating conditions in terms of quality, specifying the ethical, legal, and practical boundaries.

1. The system will observe the data protection principles of the GDPR, which include data minimization and processing data lawfully.
2. The system will function as a decision-support tool, and will not be represented as a diagnostic or treatment system.

3. The system will include an interface that is intuitive and user-friendly for the non-technical user.
4. The system will provide transparency by incorporating outputs that will explain the reasoning for the result in an interpretable manner.
5. The system will be dependable and provide reliable, consistent predictions to valid input data.
6. The system will be designed for reproducibility, allowing for experimentation and replication of results, where applicable.
7. The system will abide by the standards of professional and ethical software development, including but not limited to, documentation and maintainability.

6. Design

The proposed system design is based on the functional and non-functional requirements developed in the previous section. The system is designed to assist with the heart disease prediction respectfully (usable, ethical, and deployable) and explainable. At this point, the design is at a conceptual level and is not a detailed technical design.

The system is designed using a modular approach and can be divided into four sub-systems: data management, predictive modelling, explainable AI, and user interface. The predictive modelling sub-system supports a variety of machine learning (ML) techniques. These techniques include classical tree-based models such as Random Forest and XGBoost, as well as deep learning-based models like Artificial Neural Networks. This design enables the organization to assess and compare various models to identify the ideal model based on performance and interpretability.

An explainable AI component is integrated into the system. SHAP is used to generate global explanations of model behaviour, and LIME is used to generate local explanations. The model prediction logic and explainability are separated to promote transparency to be designed into the system, rather than to be treated as an afterthought.

Users can interact with the tool through a simple web interface where they can record a patient's details and receive predictions with explainable outputs. The tool, which is focused on supporting decisions, will not make recommendations for diagnosis or treatment.

In general, the interface design achieves a relative balance among the competing design goals of prediction, explainability and usability. The integration of machine learning, deep learning, and explainable artificial intelligence within a single framework offers a basis for developing a heart disease prediction prototype that can be considered responsible and transparent.

7. Prototype

This section describes the current prototype developed as part of the project. At this stage, the prototype focuses on the machine learning and explainability components of the system rather than a complete deployed application.

7.1. Prototype Approach

The prototyping approach adopted in this project is exploratory and incremental, with initial emphasis placed on developing and validating the predictive modelling pipeline. The current prototype consists of a fully implemented machine learning workflow, including data preprocessing, model training, evaluation, and explainability analysis.

Development began with preparation of the dataset to support reliable model training, followed by the implementation of multiple machine learning and deep learning models for comparative evaluation. This stage of the prototype was designed to identify the most suitable predictive approach in terms of both performance and interpretability.

Explainable artificial intelligence techniques were integrated into the prototype to analyse model behaviour and support transparent interpretation of predictions. At this stage, the prototype operates in an analytical environment and does not include a user-facing web interface. Integration into a web-based system is planned for a later phase of the project, following selection of the most appropriate predictive model.

7.2. Review of Technology

This review will outline the current prototype technologies with considerations for future development technologies. The rationale on the review criteria includes the technologies for experimental machine learning, the explainability, and the scope of the project.

The current prototype used Google Colab for development due to its ease of use, the computational support, and the integration with machine learning. Other local environments were considered, but rapid experimentation and reproducibility of Colab was preferred. The dataset was obtained from Kaggle (Damarla, 2024), a repository with extensive documentation available to the public.

The development of machine learning with Python, NumPy, the Pandas package, and Scikit-learn was done due to the efficiency and the stability of the frameworks with dataset structures. For explainability, the model-agnostic frameworks SHAP and LIME were employed for the global and local prediction explainability.

For future development, the web frameworks of Python, such as Flask and FastAPI, will be utilized for the backend. FastAPI was preferred because of its lightweight and REST API aspects. For the frontend development, the frameworks HTML, CSS, and the language JavaScript were employed for their ease of use. Persistent databases were considered but closed because the system works with the input of the users without the storage of the data.

7.3. Development Progress

The focus of development at this stage is the construction and verification of the machine learning component of the system. The dataset was primed and model training was supported through cleaning, feature scaling, and class balancing. Predictive models were constructed and evaluated, including Logistic Regression, Random Forest, XGBoost, and an Artificial Neural Network, and the evaluation was done using the same performance metrics. Model behaviour was analysed using XAI and SHAP for global feature importance and LIME for local, instance-level explanations. This progress sets up an operational analytical prototype and offers a validated basis for the next steps in system integration.

7.4. Reflection

This phase of the project highlights the value of structured planning and execution of the various phases required to design a data-driven healthcare system. The project has been deliberately tracked from data preparation, predictive modelling, and

explainability, to planned system integration so that analytical components could be validated before moving on to the user-facing application.

Building and assessing various machine learning and deep learning models offered valuable knowledge about the predictive behaviour with medical data, and the integration of explainable artificial intelligence (AI) promoted transparency and informed design. The consideration of system architecture and future web-based integration facilitated alignment with the primary goal of developing a useful decision support tool.

The work completed to date reflects a well-balanced emphasis on planning, analytical process development, and system level thinking, laying a robust groundwork for the next stages of application integration and user engagement.

8. Conclusion

This report focused on the potential for developing explainable machine learning systems for predicting heart disease. A literature review, systems comparison, ethical, legal, and professional consideration evaluation showed many key issues with current systems, especially addressing transparency, usability, and applicability in the real world.

The current literature suggests that while machine learning and deep learning techniques have strong predictive potential, in the health decision support systems context, they must be explainable and deployed in a responsible manner. An appropriate balance between predictive analytics and explainable, user-friendly systems design was achieved based on the user requirements and systems analysis.

The work done thus far provides a validated analytical prototype, which serves as a basis for further systems integration work. Overall, the findings support the continued development of a web-based decision support tool to enhance trust, transparency, and real world applicability in the prediction of cardiovascular disease.

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